SEP2 – UI Localization Assignment (2026)

Week 2 / AD
Description of the Exercise

Fuel Consumption and Trip Cost Calculator with Multi-Language Support

This exercise demonstrates how to build a JavaFX application that calculates fuel consumption and total trip cost with multi-language support (English, French, Japanese, and Persian). Users enter trip distance, fuel consumption rate, and fuel price. The application calculates the required fuel and total cost and displays results in the selected language.

1. Main Components and Flow

FXML Layout

- UI defined using FXML
- Includes labels, text fields, calculate button, and result display
- Language buttons (EN, FR, JP, IR) for dynamic switching

Controller Class

- Handles user interactions
- Loads language resources
- Updates UI dynamically

2. User Interface Layout

Labels:

Component ID	Description
lblDistance	Distance (km)
lblConsumption	Fuel Consumption (L/100 km)
lblPrice	Fuel Price (per liter)
lblResult	Displays results or errors

Text Fields:

Component ID	Description
txtDistance	Trip distance in kilometers
txtConsumption	Fuel consumption (L/100 km)
txtPrice	Fuel price per liter

Buttons:

Component ID	Description
btnCalculate	Calculate total fuel and cost
btnEN, btnFR, btnJP, btnIR	Language selection buttons

3. Controller Logic

Language Support

- setLanguage() loads resources based on locale
- Updates UI labels dynamically
- Displays error if resource file is missing

Calculation Logic

Total Fuel = (Consumption ÷ 100) × Distance
Total Cost = Total Fuel × Fuel Price

- Validates input
- Displays results or error message

4. Language Resource Files

messages_en_US.properties – English
messages_fr_FR.properties – French
messages_ja_JP.properties – Japanese
messages_fa_IR.properties – Persian

Example keys:

distance.label = Distance (km)
consumption.label = Fuel Consumption (L/100 km)
price.label = Fuel Price (per liter)
calculate.button = Calculate Trip Cost
result.label = Total fuel needed: {0} L | Total cost: {1}
invalid.input = Invalid input

5. Application Flow

Initialization:
- Default language: English

Language Change:
- Buttons switch locale and update UI

Calculation:
1. Enter values
2. Click Calculate
3. Validate input
4. Compute results
5. Display localized output

6. Example Workflow

Step	Action
1	User launches the application
2	Enter values: Distance 180 km, Consumption 6.5, Price 2.05
3	Click Calculate
4	Application computers and fuel and cost
5	Displays result
6	User switches language
7	UI updates to selected language

Submission Requirements

1. GitHub repository including:
- Source code
- Jenkinsfile
- Dockerfile
- Optional deployment.yaml

2. Screenshots showing all four languages (EN, FR, JP, IR)
- Include your name tag in screenshots